AGRIVIVE / DHAN-ROOT

CHHATTISGARH / PADDY BUND / TELESCOPIC OVERFLOW STANDPIPE

SERIES 05 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



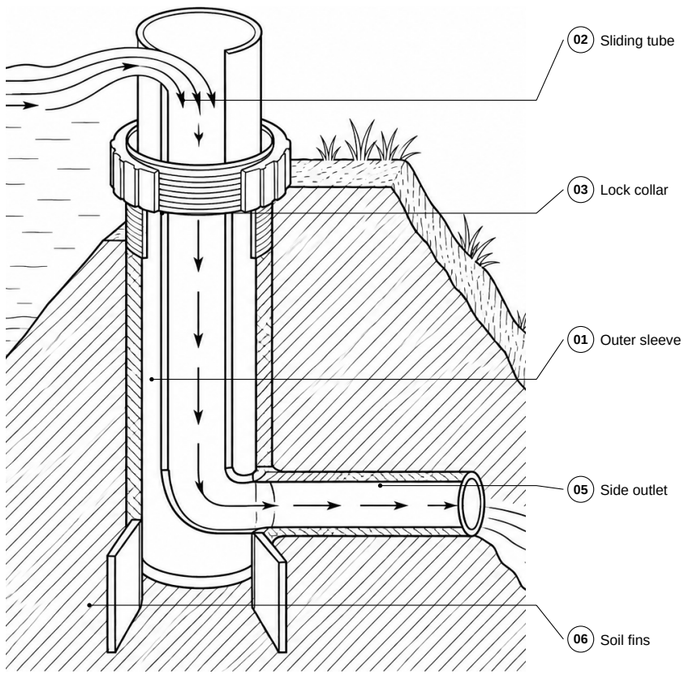
STATE CONTEXT / INDIGENOUS INSPIRATION

Rice cultivation and wide field bunds are part of Chhattisgarh's agricultural setting. DHAN-ROOT is a new adjustable overflow concept for small trial plots, not a traditional named device.

Practice source: ICAR Chhattisgarh profile | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 65 sleeve OD x 180 H

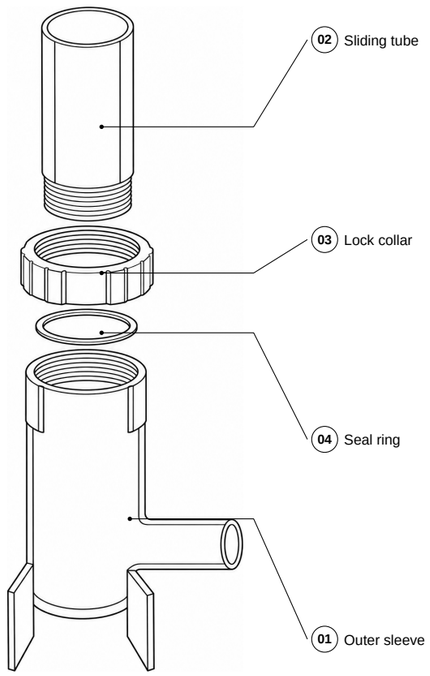


DESIGN / OPERATION NOTES

Seat the sleeve in the bund with the outlet lower than the overflow lip.
Raise or lower the open inner tube to set the nominal water level, then secure the collar.
Water spills over the open lip and travels down the bore to the side outlet.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
65 sleeve OD x 180 H

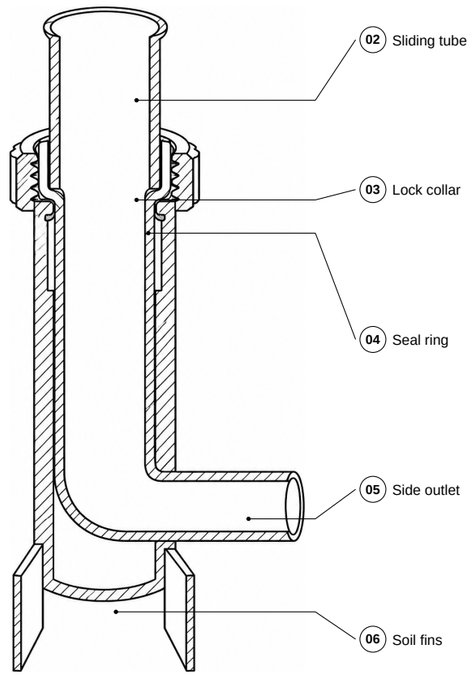


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
40 clear bore ID; 30 outlet ID; operating stroke TBD



DESIGN / OPERATION NOTES

Raise or lower the open inner tube to set the nominal water level, then secure the collar.
Water spills over the open lip and travels down the bore to the side outlet.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Sleeve OD / nominal H	**65 / 180**
Clear bore ID	**40**
Outlet ID	**30**
Operating stroke	**TBD**

02 GEOMETRIC CHECK

OUTLET BORE AREA

706.9 mm2

pi/4 x 30^2
No stage-discharge relation without tests

03 PARTS / QTY

01 Outer sleeve	**1**
02 Sliding tube	**1**
03 Lock collar	**1**
04 Seal ring	**1**
05 Side outlet	**Integral**
06 Soil fins	**Integral**

Sleeve/tube/collar: candidate polymer. Seal: compatible elastomer, grade and fit TBD.

04 INTERFACES / CHECKS

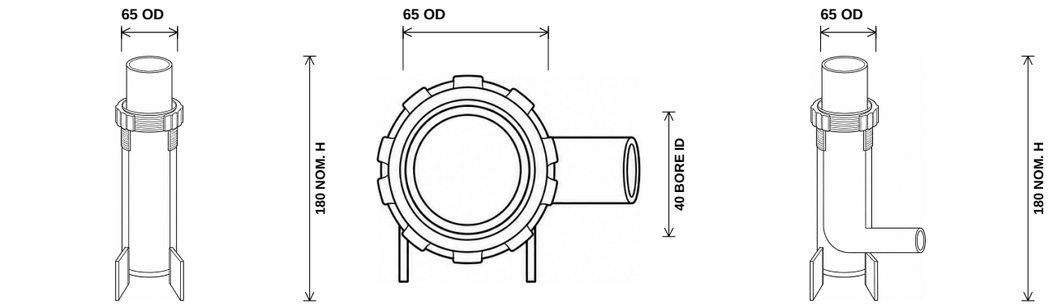
Test telescopic sealing and collar retention.

Check head, debris blockage and downstream fall.

Confirm anchorage in wet soil before field use.

PRELIMINARY / VALIDATION PENDING

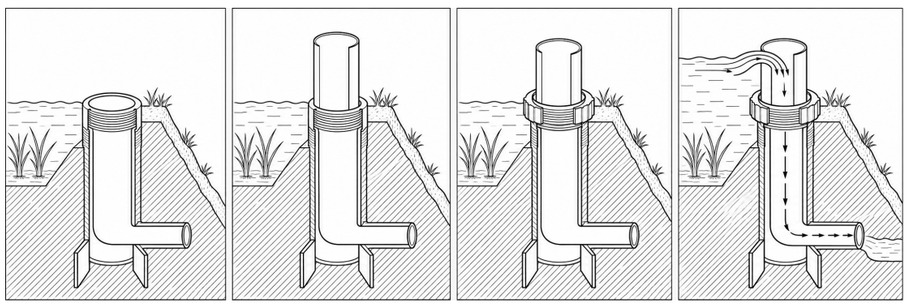
03 ORTHOGRAPHIC STUDIES / mm



FRONT / SLEEVE **TOP / SLEEVE** **SIDE / SLEEVE**

Front / side: nominal 180 H. Top: 65 sleeve OD / 40 bore ID. Outlet 30 ID.

05 INSTALLATION / OPERATING SEQUENCE



1 SET SLEEVE **2 INSERT TUBE** **3 SET / LOCK** **4 OBSERVE FLOW**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-05 / P01	DHAN-ROOT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	05 / 28